

Progressive Interface Build Visibility

Proposal: show docs and routes while a demo is still being built (typical 15–20 min builds). Draft 2026-08-12. Grounded in docs/INTERFACE_CONSTRUCTION_PLAYBOOK.md §4–§6 and observed build-timing.json samples (e.g. FTP Named Download Trigger ~17 min).

1. The problem

Today's playbook order finishes routes first, then stands up the Web UI, then generates most documentation PDFs at the end (§4 steps 8–10c). During the long middle of a build—often 10–15 minutes of route XML, transforms, and debugging—stakeholders watching the agent have little to look at. Timing data confirms routes and late “Web UI + PDFs” are the slow poles.

Goal: within the first few minutes, open a live Web UI that keeps updating as DESIGN notes, documents, and routes appear—so people can watch the interface take shape instead of waiting for a big-bang reveal at the end.

2. Current order vs proposed order

2.1 Current (§4 checklist, condensed)

- 0 Timing start → 1 DESIGN.md → 2 Module pick → 3 SQL/samples → 4 Compose
- 5–7 Routes (long pole) → 8 Web UI → 9 V2 convert → 9b–10c Docs/PDFs/tests → 11–13 Wrap-up
- Documents folder and Routes tab often stay empty until late in the session

2.2 Proposed “visibility-first” order

- **A. Minute 0–3 — Skeleton + live stage:** create demo folder, DESIGN.md stub, empty documents/, compose + Web UI copied from nearest demo / _shared, bind-mount routes + documents, **docker compose up webui** (EIP can wait), open LAN + localhost URLs
- **B. Minute 3–5 — Docs early:** drop placeholder Capability Brief stub, README skeleton, Info/Timing tabs wired to live build-timing.json; optional “Building...” banner
- **C. Minute 5+ — Routes one at a time:** author Route N V1 → convert that route to V2 immediately → Web UI Routes tab picks it up via filesystem watch / poll
- **D. Continuous:** refresh diagram PDF snapshot after each route (or on a short debounce); sync module-docs as FQCNs appear
- **E. End:** full smoke, final PDFs, finalize timing—same Definition of Done, just not the first time docs appear

3. What we would have to change

3.1 Playbook / agent process (low code risk)

Update INTERFACE_CONSTRUCTION_PLAYBOOK.md §4 so agents are required to stand up a “build stage” Web UI before the long route loop. Add explicit duties: announce URLs early; never leave documents/ empty after minute 5; convert each route to V2 as soon as that route is green enough to view (even if EIP runtime is still incomplete).

- New §4 steps: “Scaffold Web UI + open stage” before “Downstream routes”
- New required artifact: documents/BUILD_STATUS.md (or JSON) with phase, current route, last error

- Timing phases renamed to include webui_early and progressive_routes
- Hindsight retain: note when early UI helped / hurt stakeholder demos

3.2 Scaffold tooling (medium effort)

Today each demo copies webui pieces ad hoc. We need a one-command scaffold that produces a runnable “stage” UI before any real business logic exists.

- tools/scaffold_demo_stage.py — create Clients/Demos/<slug>/{webui,documents,eip-root/.../routes,tests}
- Copy route-viewer + Info + Timing from _shared / fhir-r4-platform reference
- docker-compose.yml with webui-only profile (EIP optional until first smoke)
- Seed documents/README.md, DESIGN.md stub, build-timing.json, empty diagram-groups.json
- Print both http://localhost:<port>/ and http://<LAN>:<port>/ per §1.1

3.3 Progressive documents (medium effort)

- Write DESIGN.md and BUILD_STATUS.md first; regenerate stakeholder brief as a **draft** watermarked PDF early
- export_stakeholder_brief.py --draft mode: “IN PROGRESS” banner, incomplete sections OK
- After each route: sync_module_docs.py for modules used so far (partial module-docs/ is fine)
- Debounced export_route_diagrams.py (e.g. only when route.v2.xml mtime changes; skip if no V2 yet)
- Info tab lists “available now” vs “pending” PDFs instead of waiting for a full set

3.4 Live Routes tab while building (the “watch it grow” feature)

This is the piece that matches “Web UI opens and keeps refreshing as routes are added.” It is feasible because demos already serve route.v2.xml from a bind-mounted ROUTES_DIR. We do not need to stream from the agent chat—only watch the filesystem the agent is writing.

Required Web UI / API changes

Piece	Today	Change
Routes list API	Static fetch on tab open	GET /api/v2/routes already lists files; add ?since= or ETag
Route viewer load	Manual refresh / tab switch	Poll every 3–5s while build_status.active=true
Auto-select	Often first route only	Prefer newest route.v2.xml when list grows
Build banner	None	Banner: phase, current route, last convert time, link to BUILD_STATUS
V2 convert	Batch at end	Per-route convert script hook after each V1 save
Empty state	Blank / error	Friendly “Waiting for first route.v2.xml...”

Suggested client behavior (route-viewer / app.js)

- 1) On load, fetch /api/build-status (new). If status is building, show banner and start poll loop.
- 2) Poll /api/v2/routes every N seconds; if count or mtime change, reload dropdown and diagram.
- 3) Optional: EventSource / SSE endpoint that watches routes/ with watchdog—nicer than poll, not required for v1.
- 4) Stop polling when build-status flips to complete or after idle timeout (e.g. 10 min no mtime change).
- 5) Keep Timing tab live-binding to documents/build-timing.json so phases fill in during the session.

3.5 Agent ↔ stage contract

The agent must write small, frequent artifacts the UI can read. Proposed BUILD_STATUS.json:

```
{
  "active": true,
  "phase": "routes",
  "current_route": "02-transform",
  "message": "Authoring XSLT map",
  "routes_ready": ["01-listen"],
  "updated_at": "2026-08-12T18:05:00Z"
}
```

- Update BUILD_STATUS.json at every phase change and after each successful V2 convert
- Always write route.v2.xml into the tree the Web UI mounts (or mirror immediately)
- Do not wait for “perfect” routes—viewer can show incomplete chains with a Draft badge

4. Feasibility notes

- **Possible without EIP up:** Routes tab + docs + Timing can run on Web UI alone if V2 XML is on disk
- **EIP still needed** for inject/smoke; stage UI can say “Runtime not started yet”
- **V2 convert mid-build** is already a local script pattern—just call it earlier and more often
- **Hard part:** discipline (agents must not skip early scaffold) + avoiding PDF thrash (debounce)
- **Not needed:** streaming Cursor chat into the browser; filesystem is the source of truth

5. Phased rollout

Phase	Deliverable	Effort (rough)
P0	Playbook §4 reorder + BUILD_STATUS.md convention + early compose up for next demo	0.5–1 day
P1	scaffold_demo_stage.py + webui-only profile + empty-state Routes banner	1–2 days
P2	Poll/auto-refresh Routes tab + /api/build-status + live Timing	2–3 days
P3	Draft PDF modes + debounced diagram export after each route	1–2 days
P4	Optional SSE file watcher; “Build theater” mode for sales demos	1–2 days

6. Risks and mitigations

- **Half-built routes confuse viewers** → Draft badge + BUILD_STATUS message; hide inject until EIP healthy
- **PDF regeneration slows the agent** → debounce 30–60s; skip export if no V2 change
- **Port collisions** → scaffold allocates next free Web UI port; record in DESIGN Ops
- **Agents forget early UI** → playbook hard requirement + checklist in Definition of Done
- **Stale poll after build** → auto-stop when active=false or idle

7. Success criteria

- Within ~5 minutes of “build this demo,” a stakeholder can open the Web UI on LAN
- documents/ contains at least DESIGN stub + build-timing + BUILD_STATUS before first long debug loop
- As each route.v2.xml appears, Routes tab shows it within one poll interval without manual refresh
- Total build time does not increase by more than ~10% (early convert cost offset by fewer “what’s it doing?” interruptions)
- Next three demos record webui_early phase in build-timing.json

8. Recommended next action

Implement P0 + P1 on the next net-new Sandbox demo: change the playbook order, add scaffold_demo_stage.py, and ship a Routes empty-state + 5s poll while documents/build-status.json says active. Defer fancy SSE and draft PDF watermarks until one live demo proves the theater value.

Authoritative process today: docs/INTERFACE_CONSTRUCTION_PLAYBOOK.md. This PDF is a proposal only until §4 is updated and the scaffold lands in tools/.